

A new method to forecast intermittent demand in the presence of inventory obsolescence

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Abstract: Croston is the standard method used to forecast intermittent demand items. One of its modifications, referred to as the Syntetos and Boylan approximation (SBA), has been shown in the academic literature to provide very good forecasting and inventory performance. However, Croston and SBA update demand sizes and demand intervals only in periods with positive demand, so in periods with zero demand forecasts are not adjusted downwards in the case of inventory obsolescence. The Teunter-Syntetos-Babai (TSB) method tackles this issue by updating the demand probability instead of the demand interval, and doing so in every period. This method has been shown to provide good theoretical performance for items with linear and sudden obsolescence. However, the TSB method has been shown to be empirically outperformed by the SBA method in some cases with obsolescence. In this paper, we propose a new forecasting method that is a modification of SBA. In periods with positive demand the new method updates the demand sizes, the demand intervals, and the estimator, similar to SBA, but in any time period if the actual demand interval becomes higher than the most recent estimated demand interval (which is likely to happen when the risk of obsolescence increases), the update becomes in every period similar to the probability of occurrence in TSB. The performance of the new method is analysed numerically by means of an extensive simulation experiment considering demand series with linear and sudden obsolescence and empirically by using two datasets, one coming from the military sector and the other from automotive industry. The forecast accuracy and the inventory performance are considered in the empirical investigation. The results show the outperformance of the new forecasting method in many cases dealing with obsolescence.

Keywords: Intermittent demand, forecasting, inventory obsolescence, empirical performance.

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1. Introduction

Intermittent demands are characterised by occasional demand arrivals interspersed by time intervals during which no demand occurs. Such demand patterns can characterise spare parts (e.g., engineering spares, service parts, etc.) and any stock-keeping units (SKU) within the range of products at any level of a supply chain. Such items may collectively account for up to 60% of the total stock value (Johnston et al., 2003) and are particularly prevalent in the aerospace, automotive, military, and IT sectors. Note also that stock-outs of such SKUs often imply tremendous costs, that is, very costly operational breakdowns (e.g., spare parts: aerospace, industrial production, etc.). At the same time, holding costs may be very high, especially due to the high risk of obsolescence that characterises these items (Porrás & Dekker, 2008; Saccani et al., 2017). Hence, small improvements in forecasting the demand of such SKUs may lead to substantial cost savings.

There is a plethora of methods that have been developed in the academic literature during the last decade to deal with the forecasting of intermittent demands. Two forecasting approaches have been discussed: the parametric forecasting approach and the non-parametric bootstrapping approach (Boylan & Syntetos, 2010). The former relies on a demand distributional assumption, whereas in the latter there is no distributional assumption needed to forecast the demand. Because in this paper we deal only with parametric forecasting methods, we do not focus our literature review on the bootstrapping literature but we refer the reader to Snyder et al. (2012) and Syntetos et al. (2015) for reviews of these methods.

With regard to the parametric forecasting approach, most of the research work in the area of intermittent demand is based on Croston's method. This is the standard method that is widely applied in practice and that has been largely analysed in the academic literature (Croston, 1972; Teunter et al., 2011). Croston's method was claimed to be unbiased but despite its

theoretical superiority modest benefits were recorded in the literature when it was compared with simpler forecasting techniques, such as single exponential smoothing (SES) (Willemain et al., 1994). Some empirical evidence even suggested losses in performance (Sani & Kingsman, 1997; Syntetos et al., 2015).

Syntetos & Boylan (2001, 2005) showed that Croston's method is associated with a positive bias resulting in over-forecasting mean demand. Therefore, they proposed a modification referred to as the Syntetos-Boylan approximation (SBA). SBA applies a deflating factor to Croston's method in which the factor is linear in the smoothing constant used for updating the demand intervals. SBA has been shown in a number of independent empirical studies to outperform Croston's method in terms of forecasting and inventory performance (Eaves & Kingsman, 2004; Gutierrez et al., 2008). However, some negative bias remains, which may lead in some cases to a loss of performance compared to the original Croston method (Teunter & Sani, 2009).

Furthermore, intermittent demand items are exposed, as slow movers, to a high risk of obsolescence, which makes their forecasting and stock control a challenging task. Despite this fact, there is very limited research that has discussed the issue of obsolescence in the academic literature, especially in the forecasting literature (Babai et al., 2014). It should be noted that Croston's method and SBA update demand sizes and demand intervals only in periods with positive demand, so in periods with zero demand, forecasts are not adjusted downwards considering inventory obsolescence. More recently another method, referred to as Teunter-Syntetos-Babai (TSB), has been proposed in which the demand probability is updated instead of the demand interval, doing so in every period. Hence, TSB can be used to deal with obsolescence issues by always providing up-to-date forecasts, including after long intervals with zero demand. Note that contrary to Croston's method and SBA, TSB has been shown to

be theoretically unbiased when considering an arbitrary point in time and can lead to low mean squared forecast errors. TSB has also been shown to provide good theoretical and empirical performance for items with linear and sudden obsolescence (Babai et al., 2014; Teunter et al., 2011). However, in some situations the TSB method has been shown to be empirically outperformed by the SBA method, which is due to the update in every period of the demand probability that may affect the mean squared error (MSE) performance of the estimator, especially for non-stationary demands. The performance of SES has also been analysed in the academic literature, and it has been shown that this method leads to a bias reduction when it estimates intermittent demands, and it may outperform Croston's method and SBA when real data are used with a decreasing demand pattern. This good performance of SES is attributed to its update of the forecasts in every period, which enables adjusting them downwards in the presence of obsolescence (Babai et al., 2014). Recently, a new Croston-type method has been proposed by Prestwich et al. (2014) to deal with intermittent demand and the issue of obsolescence. The method, referred to as hyperbolic-exponential smoothing (HES), combines exponential smoothing with an application of Bayesian inference when no demand occurs. Through a numerical investigation with generated data, HES has been shown to lead to high forecasting accuracy for intermittent and non-intermittent demand and to outperform TSB in some cases. However, no inventory performance has been analysed for this method.

In this paper, we propose a new forecasting method that is a modification of the SBA estimator. In periods with positive demand, it updates the demand sizes, demand intervals, and the estimator, similar to SBA, but at any time period if the actual demand interval becomes higher than the most recent estimated demand interval (which is likely to happen in cases of obsolescence), the update of the demand interval becomes as is done for the probability of occurrence in the TSB estimator (i.e., in every period). Hence, the new method

takes advantage of TSB by capturing the risk of obsolescence and updating in every period to reduce the bias and takes advantage of SBA by using the same estimator when there is no obsolescence, which has also shown a high MSE empirical performance even under non-stationary demands. The performance of the new method is assessed numerically by means of an extensive simulation experiment considering generated demand series with linear and sudden obsolescence and empirically by using two datasets, one of which comes from the military sector and the other the automotive industry. Forecast accuracy and inventory performance are considered in the empirical investigation. The empirical performance is analysed as well by means of datasets of selected SKUs that are characterised by decreasing demand patterns.

The remainder of the paper is organised as follows. Section 2 presents the benchmark forecasting methods considered in our paper along with the new proposed method. We provide in Section 3 the details of the numerical investigation. The simulation data and settings are presented followed by the numerical results. Section 4 is dedicated to the empirical investigation of the forecast accuracy and the inventory performance of the forecasting methods. The empirical results are shown when complete datasets are used and when selected SKUs have a decreasing demand. The conclusions and some avenues for further research are provided in Section 5.

2. Forecasting methods

In this section, after presenting some notations needed in the paper, we provide the expression of the benchmark forecasting methods that we consider in our numerical and empirical investigations. We then present our new proposed method.

2.1. Notation and benchmark forecasting methods

The following notations will be used throughout the paper:

D_t : Demand for an item at time t

D_t' : Estimate of mean demand per period made at time t for period $t+1$

Z_t : Actual demand size at time t

Z_t' : Estimate of the demand size at time t

T_t : Actual demand interval at time t

T_t' : Estimate of the demand interval at time t

p_t' : Estimate of the demand occurrence probability at time t

α, β : Smoothing parameters ($0 \leq \alpha \leq 1, 0 \leq \beta \leq 1$)

SES updates the forecast of the demand at the end of every period as follows:

$$D_t' = D_{t-1}' + \alpha(D_t - D_{t-1}') = \alpha \sum_{i=0}^{\infty} (1 - \alpha)^i D_{t-i}.$$

The Croston's method estimates separately demand sizes and demand intervals, which are updated only in periods with positive demand by means of simple exponential smoothing as follows:

$$D_t' = \frac{Z_t'}{T_t'}$$

$$\text{where } \begin{cases} \text{If } D_t > 0: Z_t' = Z_{t-1}' + \alpha(Z_t - Z_{t-1}') \text{ and } T_t' = T_{t-1}' + \beta(T_t - T_{t-1}') \\ \text{Otherwise : } Z_t' = Z_{t-1}' \text{ and } T_t' = T_{t-1}' \end{cases}$$

Note that in all the expressions of the forecasting methods described in this section, if $D_t > 0$ then $Z_t = D_t$. We also note that Croston originally suggested using the same value for both smoothing constants, but in this paper we use two different smoothing constants to test the performance of this method under better conditions, as initially proposed by Schultz (1987).

SBA uses separate exponentially smoothed estimates of demand sizes and demand intervals. It relies on the employment of a correction factor $(1 - \beta/2)$ to the estimates produced by Croston's method. The estimator is given by:

$$D_t' = (1 - \frac{\beta}{2}) \frac{Z_t'}{T_t'}$$

$$\text{where } \begin{cases} \text{If } D_t > 0: Z_t' = Z_{t-1}' + \alpha(Z_t - Z_{t-1}') \text{ and } T_t' = T_{t-1}' + \beta(T_t - T_{t-1}') \\ \text{Otherwise : } Z_t' = Z_{t-1}' \text{ and } T_t' = T_{t-1}' \end{cases}$$

The TSB method uses separate exponentially smoothed estimates of the demand probability and the demand size. The estimate of the probability of occurrence is updated in every time period. The estimate of the demand size is updated only at the end of periods with positive demand.

$$D_t' = p_t' Z_t'$$

$$\text{where } \begin{cases} \text{If } D_t > 0: Z_t' = Z_{t-1}' + \alpha(Z_t - Z_{t-1}') \text{ and } p_t' = p_{t-1}' + \beta(1 - p_{t-1}') \\ \text{Otherwise : } Z_t' = Z_{t-1}' \text{ and } p_t' = p_{t-1}' + \beta(0 - p_{t-1}') \end{cases}$$

Note that only these methods are considered as benchmarks in this paper because numerical and empirical results are available for these methods in the literature under the same settings ($\alpha = \beta$, $\alpha \neq \beta$) and demand data (theoretical data with linear and sudden obsolescence, RAF and automotive datasets) as shown in Table 1, which enables a better linkage of our findings with previous research. Hence, in addition to the performance evaluation of the modified SBA, all the results using the benchmark methods are also replicated in this paper.

Method	Forecast accuracy results				Inventory performance results	
	Numerical results		Empirical results			
	$\alpha = \beta$	$\alpha \neq \beta$	RAF dataset	Automotive dataset	RAF dataset	Automotive dataset
SES			Teunter & Duncan (2008)	Syntetos and Boylan (2005)	Syntetos & Boylan (2006)	Teunter & Duncan (2008)
Croston	Syntetos & Boylan (2001)	Teunter et al. (2011)	Teunter & Duncan (2008)	Syntetos and Boylan (2005)	Syntetos & Boylan (2006)	Teunter & Duncan (2008)
SBA	Syntetos & Boylan (2005)	Teunter et al. (2011)	Teunter & Duncan (2008)	Syntetos and Boylan (2005)	Syntetos & Boylan (2006)	Teunter & Duncan (2008)
TSB	Teunter et al. (2011)	Teunter et al. (2011)	Babai et al. (2014)	Babai et al. (2014)	Babai et al. (2014)	Babai et al. (2014)
Modified SBA	Current paper					

Table 1. Methods considered in the paper and related results

2.2. New forecasting method

We propose a new forecasting method that is a modification of the SBA method. The method will be referred hereafter to as the ‘modified SBA’ method. In periods with positive demand, the modified SBA method updates demand sizes, demand intervals, and the estimator exactly like the SBA method, but in any period (with zero demand), if the actual demand interval becomes higher than the most recent estimated demand interval, the update becomes in every period similar to the probability of occurrence in the TSB method. The estimator of the modified SBA method is given by:

$$D_t' = (1 - \frac{\beta}{2}) \frac{Z_t'}{T_t'}$$

$$\text{where } \begin{cases} \text{If } D_t > 0: Z_t' = Z_{t-1}' + \alpha(Z_t - Z_{t-1}') \text{ and } T_t' = T_{t-1}' + \beta(T_t - T_{t-1}') \\ \text{Otherwise: } \begin{cases} T_t' = \begin{cases} T_{t-1}' + \beta(T_t - T_{t-1}') & \text{If } T_t > T_{t-1}' \\ T_{t-1}' & \text{If } T_t \leq T_{t-1}' \end{cases} \\ Z_t' = Z_{t-1}' \end{cases} \end{cases}$$

If the probability of obsolescence increases, the demand intervals tend to increase, which can be captured by the modified SBA method, because the update of the estimation of the intervals becomes in every period, implying therefore an adjustment downwards of the forecasts. Obviously, this downward adjustment when the risk of obsolescence increases cannot be captured by Croston and SBA, which only update demand sizes and intervals after demand occurrences. However, TSB and SES can do so due to their update in every period. This behaviour of the two methods will be analysed hereafter in the paper by considering a context of intermittent demands associated with a risk of obsolescence.

In the following section, the performance of the five forecasting methods is analysed through a numerical investigation.

3. Numerical investigation

3.1 Data and simulation settings

For the purpose of our numerical investigation, we generate demand data for a 10-year demand history (i.e., 120 months/periods). The demand sizes are assumed to be stationary, and they are generated from a logarithmic distribution with the shape parameter c ($0 < c < 1$). The demand occurrences are generated from the Bernoulli distribution with parameter p ($0 < p < 1$). The logarithmic and Bernoulli distributions have considerable empirical

evidence in their support to model demand sizes and occurrence probability, respectively, and they have been recommended in the academic literature (e.g., Kwan, 1991).

In order to generate the demand sizes, two cases of the logarithmic parameter are considered, namely, $c = 0.001$ and $c = 0.9$. This is equivalent to generated demand sizes with means and standard deviations equal to $\mu = 1$, $\sigma = 0$ and $\mu = 4$, $\sigma = 5$, respectively.

In the former case, this results in a very slow-moving demand (with unit-sized transactions), whereas in the latter case, demand patterns with higher variability are generated.

In order to generate the demand occurrences while taking into account the obsolescence issue, we assume two cases: (1) a linearly decreasing demand and (2) a demand with a sudden obsolescence. The numerical parameters of the Bernoulli-generated demand are as follows. In linearly decreasing demand, the demand probability starts at $p_0 = 0.2$ or $p_0 = 0.5$ in the first period and decreases linearly to 0 in the last period. In sudden obsolescence, we assume that after five years of stationary demand with demand probability $p = 0.2$ or $p = 0.5$, the demand probability suddenly drops to 0.

An additional case with a lower demand occurrence probability $p_0 = 0.1$ and a high demand size distribution parameter $c = 0.9$, which corresponds to very lumpy demand patterns, will also be generated to evaluate the performance of the methods under this specific situation.

Note that in order to deal with the sampling error when generating the demand sizes and occurrences, 10 replications of the demand history data are generated for each of the control parameter combinations.

The smoothing constants used to update the demand in SES and the demand sizes, intervals, and probabilities in the Croston's method, SBA, TSB, and the modified SBA are varied from 0.05 to 0.3 with a step increase of 0.05. For TSB, note that for a better tuning of the estimator we also include low values of the smoothing constant of the demand occurrence probability

ranging from 0.01 and 0.05 with a step increase of 0.01. The first period of each demand series is used to initialise the estimates (by using the theoretical mean demand), then the entire demand history is used to evaluate the performance of the forecasting methods. The settings described in this subsection facilitate the reproduction (Boylan, 2016) of our results should one wish to do so.

The performance of the five considered forecasting methods is evaluated by means of two accuracy measures, namely, the mean error (*ME*) and the mean square error (*MSE*). The forecast error of each SKU is calculated first then the errors are averaged across SKUs. The *ME* is a measure that reflects the bias (and its direction) of the forecasting method, whereas the *MSE* is a measure that expresses the variance of the forecast errors and the bias. For more details about these two measures, the reader is referred to Babai et al. (2014). Obviously, other measures could have been used to assess the performance of the forecasting methods, such as the mean absolute scaled error (*MASE*; Hyndman & Koehler, 2006), the *sMSE* measure proposed in Petropoulos and Kourentzes (2015), and the mean-based *MSE* discussed in Prestwich et al. (2014). However, to allow for a validation and comparison of the findings with those presented in Teunter et al. (2011) and Babai et al. (2014) and to simplify the presentation of the numerical results in the paper, we opt to use only the *ME* and *MSE* measures as accuracy measures, and we include, in addition, inventory performance measures. The inventory performance will be measured in the paper through the inventory holding and backordering volumes resulting from the use of the different forecasting methods.

3.2. Numerical results

In order to avoid the presentation of the performance results of all the considered forecasting methods under the high combinatorial of the selected smoothing constants, we opt in this

paper to analyse the average performance of the methods when the smoothing constants are selected on the basis of providing the smallest MSE averaged across all SKUs.

Table 2 reports the ME and MSE results of the five forecasting methods under the four combinations of the demand size and demand probability distribution parameters. Note that Table 2 shows the average performance when the smoothing constants are selected on the basis of providing the smallest MSE (over the out-of-sample) averaged across all demand series. The optimal smoothing constants are also reported.

Forecasting method		SES	Croston	SBA	TSB	Modified SBA
$c = 0.001$ $p = 0.2$	α	0.05	0.05	0.05	0.05	0.05
	β		0.15	0.25	0.2	0.05
	ME	0.0285	0.0735	0.0496	0.0331	0.0227
	MSE	0.0929	0.0974	0.0952	0.0926	0.091
$c = 0.001$ $p = 0.5$	α	0.05	0.05	0.05	0.05	0.05
	β		0.2	0.25	0.25	0.1
	ME	0.0703	0.101	0.0577	0.0702	0.0123
	MSE	0.1674	0.1761	0.1694	0.1674	0.1641
$c = 0.9$ $p = 0.2$	α	0.05	0.15	0.1	0.1	0.05
	β		0.15	0.3	0.2	0.1
	ME	0.1159	0.2814	0.1925	0.1647	0.0938
	MSE	2.7861	2.8319	2.7847	2.7667	2.7335
$c = 0.9$ $p = 0.5$	α	0.05	0.05	0.05	0.05	0.05
	β		0.15	0.2	0.3	0.1
	ME	0.2695	0.4755	0.317	0.1842	0.0679
	MSE	7.3247	7.5502	7.4344	7.2699	7.2076

Table 2. Results for a linearly decreasing demand

Table 2 shows a positive bias resulting from all the forecasting methods, which means that none of these five methods directly incorporates the downward trend in demand. In addition, the results also show that Croston's method is associated with the highest bias, whereas TSB and SES provide lower biases, which confirms what has been stated in Teunter et al. (2011). This result is expected because these two latter estimators have been shown in the academic literature to be theoretically unbiased when all points in time are considered. Table 2 shows

that the modified SBA leads the lowest bias with a bias reduction compared to that of SES and TSB (the second least-biased methods) going up to 80% when the parameters of the demand size and probability distributions are $c = 0.001$ and $p = 0.5$.

By looking at the MSE results, Table 2 shows that Croston's method leads to the worst performance, whereas the modified SBA outperforms all the other methods with an MSE reduction that goes up to 2% when compared to TSB, being the method with the second best MSE results. Note, however, that except for Croston's method, all forecasting methods are associated with a very close MSE performance.

Hence, it should be noted that overall, the modified SBA method results in the best performance in terms of bias and MSE followed by TSB and SES. This is expected because these three methods are the only ones that update the demand not only in periods with a positive demand, leading to an adjustment in the decrease in demand and a better capture of the obsolescence.

The results in Table 3 are in line with those presented in the previous case. In fact, the new method is the best-performing method in terms of ME and MSE, followed by TSB and SES. However, the relative performance differences of SBA and Croston are much larger than those reported for the previous case with the linearly decreasing demand. In fact, the new method reduces, for example, the ME and the MSE when compared to SBA by 92% and 35%, respectively, when the parameters of the demand size and probability distributions are $c = 0.001$ and $p = 0.5$. The reduction in the ME and the MSE when compared to Croston are 94% and 42%, respectively. This is expected because Croston and SBA are unable to react to sudden obsolescence, whereas the occurrence of positive demands did allow adjustment, albeit at a slow pace, for the linearly decreasing demand. It is obvious therefore that the new method outperforms Croston or SBA after the occurrence of the sudden obsolescence.

Forecasting method		SES	Croston	SBA	TSB	Modified SBA
$c = 0.001$ $p = 0.2$	α	0.05	0.05	0.05	0.05	0.05
	β		0.05	0.3	0.25	0.1
	ME	0.0321	0.1072	0.0876	0.0321	0.013
	MSE	0.0806	0.0956	0.0939	0.0806	0.0797
$c = 0.001$ $p = 0.5$	α	0.1	0.3	0.3	0.05	0.05
	β		0.1	0.3	0.3	0.15
	ME	0.042	0.2519	0.1851	0.0415	0.0142
	MSE	0.1439	0.2478	0.2198	0.1439	0.1417
$c = 0.9$ $p = 0.2$	α	0.05	0.05	0.05	0.05	0.05
	β		0.05	0.05	0.3	0.1
	ME	0.1251	0.4394	0.4195	0.0807	0.0706
	MSE	2.908	3.1292	3.111	2.8497	2.8368
$c = 0.9$ $p = 0.5$	α	0.1	0.05	0.05	0.05	0.05
	β		0.05	0.05	0.3	0.15
	ME	0.1639	1.039	0.9893	0.189	0.0645
	MSE	7.3735	9.076	8.9663	7.1805	7.1671

Table 3. Results for a demand with sudden obsolescence

The detailed ME and MSE results for the example of a very lumpy demand case (with a linear decreasing demand) are reported in Table A1 in Appendix A. These results show that when very lumpy demand patterns are considered, SBA outperforms the TSB method in many cases by resulting in lower MSE results, which is in line with the empirical findings of Babai et al. (2014). It should be noted that the modified SBA remains the best-performing method in these cases.

4. Empirical investigation

In this section, we discuss details related to our empirical investigation. First, we describe the empirical data used in this research as well as the experimental structure and some simulation-related details. Second, we provide the results of our empirical investigation.

4.1. Empirical data and settings

Two datasets are used for the purpose of validating the findings of our numerical investigation. The first dataset comes from the Royal Air Force (RAF, UK), and it consists of

the individual monthly demand over a period of seven years (84 observations) for 5,000 SKUs. The second dataset is a monthly demand history of 3,000 SKUs over a period of two years (24 observations) from the automotive industry. These datasets are the same used in the empirical investigations conducted by Syntetos & Boylan (2006), Teunter et al. (2011), and Babai et al. (2014).

Descriptive statistics (across all SKUs) of dataset 1 and dataset 2 are provided in Table 4 and Table 5, respectively. The mean and standard deviation of the demand sizes, the demand intervals, and the demand per period are calculated for every single SKU/demand history. Subsequently, the minimum, 25th percentile, median, mean, 75th percentile, and maximum across all SKUs are presented. The descriptive statistics are rounded to three decimal places.

5,000 SKUs	Demand intervals		Demand sizes		Demand per period	
	Mean	St. Dev.	Mean	St. Dev.	Mean	St. Dev.
Min	3.824	0.000	1.000	0.408	0.036	0.187
25%ile	7.273	5.431	1.556	2.964	0.155	0.538
Median	9.000	6.930	3.833	7.203	0.369	1.452
75%ile	11.571	8.630	11.333	14.328	1.155	4.434
Max	24.000	16.460	668.000	1138.408	65.083	275.706

Table 4. Descriptive statistics of dataset 1

3,000 SKUs	Demand intervals		Demand sizes		Demand per period	
	Mean	St. Dev.	Mean	St. Dev.	Mean	St. Dev.
Min	1.043	0.209	1.000	0.000	0.542	0.504
25%ile	1.095	0.301	2.050	1.137	1.458	1.319
Median	1.263	0.523	2.886	1.761	2.333	1.922
75%ile	1.412	0.733	5.000	3.357	4.167	3.502
Max	2.000	1.595	193.750	101.415	129.167	122.746

Table 5. Descriptive statistics of dataset 2

Note that the RAF dataset 2 consists of SKUs with a very high degree of lumpiness. The standard deviation of the demand intervals varies between 0 and 16.46 with a median equal to 6.93. The standard deviation of the demand sizes varies between 0.408 and 1138.4 with a median equal to 7.2. From Table 5, the dataset coming from the automotive industry contains SKUs with a low degree of intermittence. The standard deviation of the demand intervals varies between 0.2 and 1.6, whereas the standard deviation of the demand sizes varies between 0 and 101 with a median equal to 1.7. Some SKUs in each dataset are characterised by a decreasing demand over the history. The data related to those SKUs will be described in more detail in the following subsection.

We simulate using real data the performance of the five forecasting methods. We split the demand history available for each SKU into two parts. The first part (i.e., within-sample) is used in order to initialise the estimates of the level and variance of demand. This part is composed of 36 periods in dataset 1 and in 13 periods in dataset 2. Exponential smoothing is initialised based on the average demand over the initialisation block. For Croston's method, SBA, and the modified SBA, an initial value is required for the demand size and demand interval. Those are also obtained as averages over the initialisation block. For TSB, the initial estimates of the demand sizes and demand occurrence probability are also obtained through averages over the initialisation block. The forecasts are then calculated in the second part (out-of-sample), which is used for the evaluation of performance.

4.2. Empirical results on forecast accuracy

We start this section by presenting the forecast accuracy results of the five methods when the complete datasets are used. Tables 6 and 7 show the average performance when the smoothing constants are selected on the basis of providing the smallest MSE averaged across

all SKUs. The optimal smoothing constants are also reported. Detailed results for all considered values of the smoothing constants are given in Appendix B.

Forecasting method	α	β	ME	MSE
SES	0.05		-0.118	233.804
Croston	0.15	0.2	0.414	232.906
SBA	0.15	0.3	0.156	230.961
TSB	0.15	0.02	-0.178	232.202
Modified SBA	0.15	0.5	-0.076	229.993

Table 6. Empirical results for dataset 1

Forecasting method	α	β	ME	MSE
SES	0.05		-0.066	74.970
Croston	0.05	0.1	0.116	75.137
SBA	0.1	0.1	-0.107	74.923
TSB	0.05	0.05	-0.064	75.033
Modified SBA	0.1	0.1	-0.134	74.899

Table 7. Empirical results for dataset 2

The results in Tables 6 and 7 show that the modified SBA outperforms all the methods in terms of the ME and MSE followed by SBA. By looking at the details of the results in each dataset, because the demands are associated with a high degree of lumpiness for dataset 1, the optimal smoothing constants of the demand sizes and intervals are relatively large, which is expected due to the non-stationarity of the demand, as has been stated in Babai et al. (2014). Moreover, Croston's method leads to a very poor performance with a high positive bias and MSE, whereas SES and TSB result in almost the same bias and MSE results. SBA performs relatively well because it leads to a very close performance to that of the new method, especially in terms of the MSE. Note that SBA is the only method that leads to a negative bias in this dataset 1, which is in line with what has been reported in the academic literature.

The results in Table 7 show that lower optimal smoothing constants are used compared to the previous case, which is due to the lower lumpiness and the stationarity of the demand in

dataset 2. This also results in SES and TSB having the lowest bias, which confirms what has been shown in the numerical investigation. This is also theoretically expected because these two methods are unbiased when all points in time are considered. However, SBA and the modified SBA lead to the best MSE results, whereas Croston's method provides the highest MSE. Note that in this dataset, similar to SBA, the new method leads to a negative bias that is higher than that of SES and TSB. It should also be noted that in dataset 2 the difference in the ME and MSE results is overall very low, which can be explained by the low degree of intermittence of the demand in this dataset.

We have also evaluated the forecast accuracy of the methods when an MSE-based optimisation per SKU (with an ex ante approach) is considered. Table 8 shows the ME and MSE results for dataset 1 along with the average optimal smoothing constants. Note that the short length of the demand history dataset 2 does not enable generating MSE-based optimisation per SKU (with an ex ante approach). It is important to note that an ex post methodology could have also been considered for the purposes of our experiments (see, for example, Porras & Dekker, 2008), but we think that the ex ante approach is preferred because it enables the simulation of the performance in a 'dynamic' way, that is, what would have happened if the methods under consideration were used in practice.

Estimator	α	β	ME	MSE
SES	0.061		0.024	265.975
Croston	0.177	0.152	0.318	266.924
SBA	0.181	0.163	0.153	264.333
TSB	0.181	0.041	-0.063	262.057
Modified SBA	0.180	0.144	-0.039	263.355

Table 8. Minimum (per SKU) MSE results – dataset 1

The results in Table 8 differ from those presented in Table 6 in the sense that when the optimisation is done per SKU, TSB and modified SBA lead to the best performance. The former results in a lower MSE compared to the latter, but it is characterised by a higher ME.

In order to better analyse the empirical comparative performance of the methods for SKUs with a high inventory obsolescence risk and to show that the modified SBA performs well for such SKUs, we have identified and separated in each dataset the SKUs that are associated with a decreasing demand pattern. The identification process is as follows:

- For dataset 1, we split the demand history into four equal non-overlapping blocks in which each block consists of 21 periods. We then calculate the mean demand over each block. The SKUs related to a strict decrease of the mean from the older time block to the most recent were identified.
- In dataset 2, we apply the same procedure as for dataset 1 with the only difference that the demand history is divided into only three equal non-overlapping blocks of 8 periods. Note that fewer blocks are used in this dataset due to the relatively shorter demand history.

The selection procedures just described result in 190 SKUs for dataset 1 and 469 SKUs for dataset 2. Those selected SKUs are used to analyse the comparative performance of the forecasting methods under a decreasing demand pattern and to validate the findings of the numerical investigation.

We present in Tables 9 and 10 the performance results (minimum, across SKUs, MSE results) of the five methods and the optimal smoothing constants for dataset 1 and dataset 2, respectively.

Forecasting method	α	β	ME	MSE
SES	0.05		-1.269	60.566
Croston	0.3	0.3	1.931	67.248
SBA	0.3	0.3	1.500	61.323
TSB	0.3	0.05	-1.345	61.310
Modified SBA	0.3	0.3	1.027	58.649

Table 9. Minimum (across SKUs) MSE results for SKUs with decreasing demand – dataset 1

Forecasting method	α	β	ME	MSE
SES	0.25		-0.903	35.151
Croston	0.3	0.15	1.074	35.113
SBA	0.2	0.3	0.541	31.271
TSB	0.05	0.15	-0.951	34.825
Modified SBA	0.2	0.3	0.483	31.168

Table 10. Minimum (across SKUs) MSE results for SKUs with decreasing demand – dataset 2

The results in Tables 9 and 10 show that when SKUs with only a decreasing demand are considered, the new method outperforms SBA and TSB in terms of bias and MSE. In this case, again Croston's method leads to the worst bias and MSE performance. The new method reduces the bias of Croston's method by 46% and the MSE by 12% in the RAF dataset. In the automotive dataset, the new method results in 55% and 11% reduction in the bias and MSE, respectively, as compared to that of Croston's method. As expected, the optimal smoothing constants in this case are relatively larger than that when the complete datasets are considered, which is obviously due to non-stationarity and the higher variability of the demand of the SKUs in these selected demand data.

4.3. Empirical results on inventory performance

In order to evaluate the inventory performance of the forecasting methods, we consider an order-up-to-level policy, that is, the inventory position is assessed at the end of every review period and a demand arrival will subsequently trigger a replenishment order that will allow the inventory position to rise up to the level S . We assume that if the demand cannot be satisfied directly from stock, the relevant quantity is backordered (Babai et al., 2014; Halkos et al., 2018).

By using the forecast of the demand and the smoothed MSE at each period t , denoted by D_t' and MSE_t' , respectively, the lead-time demand can be estimated. The optimal order-up-to-

level is then calculated at each period as $(L+1)D'_t + k\sqrt{(L+1)MSE'_t}$, where L is the lead time and k the safety stock factor. Descriptive statistics (across SKUs) on lead times in datasets 1 and 2 are provided in Table 11.

	Dataset 1	Dataset 2
Min	1	1
25%ile	5	1
Median	9	1
75%ile	12	1
Max	33	1

Table 11. Lead-time information

We vary the safety factor from 1 to 4 in order to calculate the resulting holding and backordering volumes under the five forecasting methods. In order to make fair comparisons among the methods, despite the fact that they do not result in exactly the same service performance (for the same value of the safety factor), we draw efficiency curves of holding versus backordering. The efficiency curves obtained for datasets 1 and 2 are reported in Figures 1 and 2, respectively. These efficiency curves can be interpreted as signifying that for a certain inventory holding volume, the method that has its curve closer to the x -axis implies a lower backordering volume and thus more efficiency.

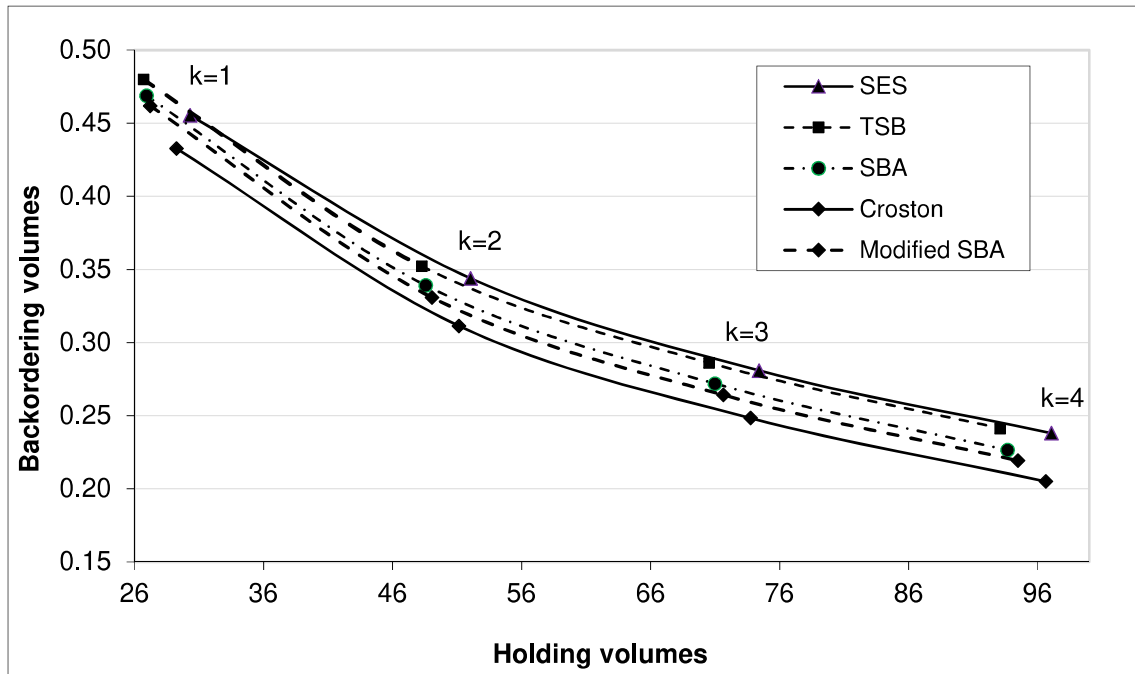


Figure 1. Stock control efficiency curves – dataset 1

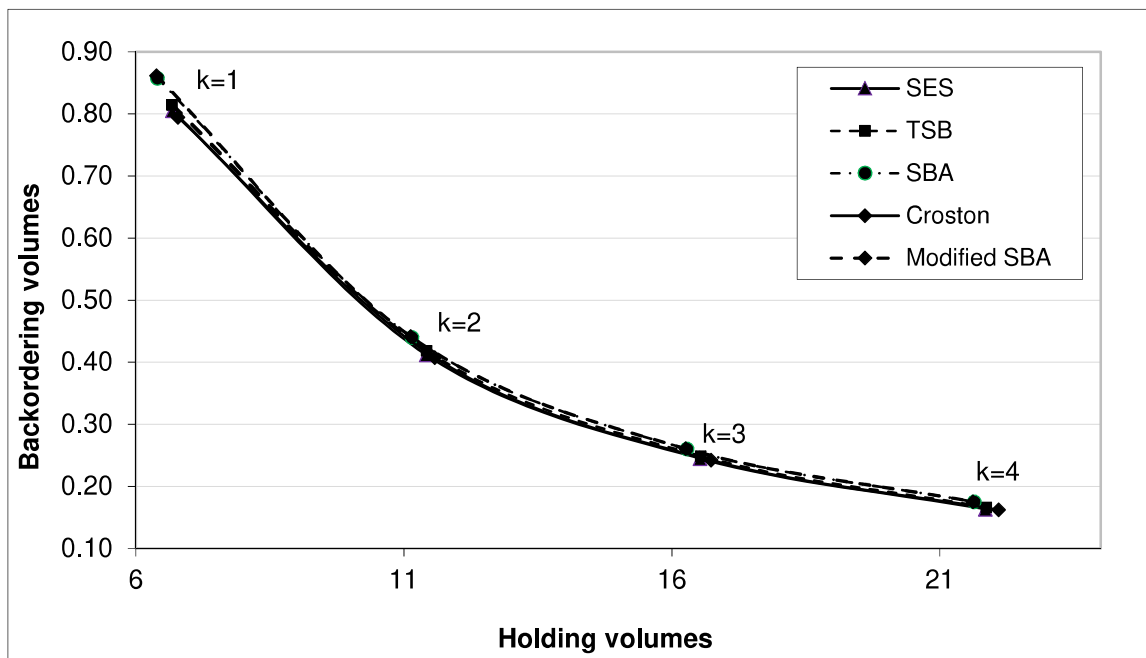


Figure 2. Stock control efficiency curves – dataset 2

Figure 1 shows that for dataset 1, Croston's method outperforms all other methods, which confirms what has been shown in Babai et al. (2014) regarding its outperformance to SBA, TSB, and SES. Note that the second-best performance is obtained with the modified SBA

method. The results also show that SBA outperforms TSB, which is in line with what has been shown for the empirical forecast accuracy of these two methods in Section 4.2. Figure 2 shows that all the methods result in a similar inventory performance because their efficiency curves almost overlap. This is in line with the small difference in the forecast accuracy among the methods shown in Section 4.2.

We also report in Figures 3 and 4 the results obtained after the selection of SKUs with decreasing mean demand.

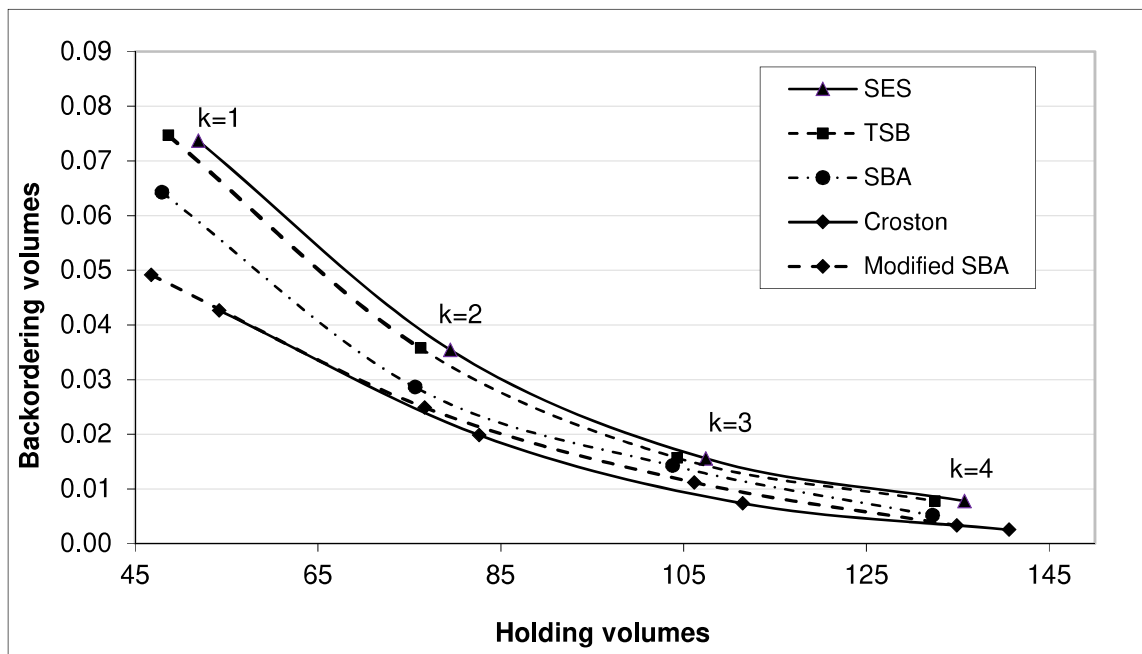


Figure 3. Stock control efficiency curves for SKUs with decreasing demand – dataset 1

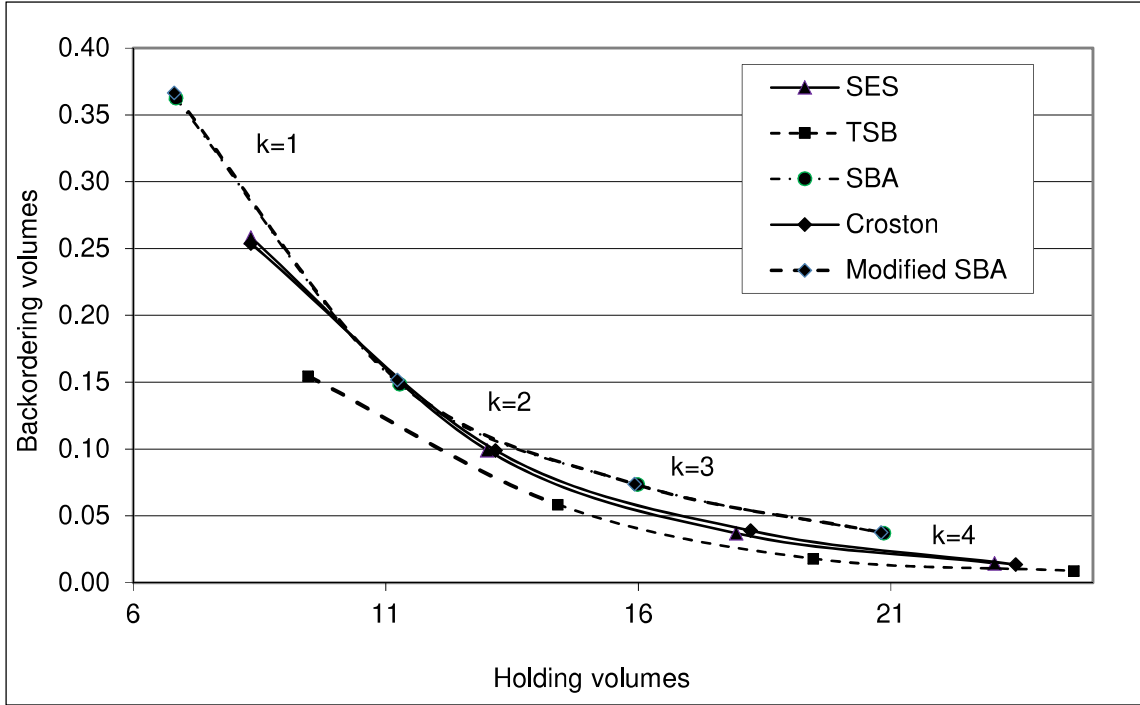


Figure 4. Stock control efficiency curves for SKUs with decreasing demand – dataset 2

When the SKUs with decreasing demand are selected in dataset 1, the performance of the modified SBA improves compared to Croston's method, and these two methods still outperform all the other methods. However, contrary to dataset 1, the results in dataset 2 show a clear outperformance of TSB compared to all the other methods. It should be noted that in dataset 2 the SKUs are associated with low-average demand intervals, which means that these SKUs reflect a fast intermittent demand so the modified SBA coincides with SBA.

5. Conclusions

Obsolescence is a very important issue that is increasingly challenging inventory managers of intermittent demand SKUs in many sectors and particularly in spare parts management. This is due to the high costs associated with this issue, including inventory-holding costs, scrapping costs, and so on. Despite this fact, there is a relatively limited discussion of obsolescence in the academic literature, especially in the forecasting literature (Babai et al., 2014; Prestwich et al., 2014; Teunter et al., 2011). This paper attempts to bridge this gap by

proposing a new forecasting method, referred to as the ‘modified SBA’, which deals with intermittent demand patterns that are characterised by a high risk of obsolescence.

In the modified SBA method, unless the actual demand interval in any period is higher than the estimated demand interval, the updates of the demand sizes and demand intervals are performed in periods with positive demands only. This updating mechanism allows for an adjustment downwards of the forecasts when the risk of obsolescence increases, which can be captured by SES and TSB as well because they are both characterised by an updating mechanism in every period either for the estimate of the demand per period or for the demand intervals. Obviously, this cannot be captured by Croston and SBA, which update demand sizes and intervals only after demand occurrences.

We have conducted a numerical investigation in this paper to analyse the forecast accuracy of the five forecasting methods when two generated demand patterns are considered, namely, a linearly decreasing demand and a demand with sudden obsolescence. The numerical results show that when bias and MSE measures are considered, the new method is the best-performing method, followed by TSB. The new method has been shown to lead to a substantial bias reduction that can reach 80% and an MSE reduction of up to 2% when compared to TSB (the second-best-performing method). The results also show that in obsolescence with theoretically generated data, Croston’s method and SBA do not perform well, which is expected because these two methods are unable to capture the risk of obsolescence by updating the forecasts only after demand occurrences.

The empirical results show that the modified SBA outperforms all the methods in terms of ME and MSE. The results differ from those obtained in the numerical investigation because, overall, the second-best-performing method is SBA and not TSB. This is in line with what has been shown in Babai et al. (2014) with regards to the good empirical performance of SBA. The results also show that Croston’s method leads to very poor forecast accuracy

when empirical data are used. Moreover, when only the selected SKUs associated with a decreasing demand are used in the empirical investigation, the results show a larger benefit of using the modified SBA compared to when complete datasets are used. The results also show that the optimal smoothing constants used, when SKUs with a decreasing demand are considered, are relatively larger than that when complete datasets are considered, which is obviously due to the non-stationarity of the demand of the SKUs in these selected demand data.

The analysis of the inventory performance reveals that, for the dataset with low degrees of intermittence, the different forecasting methods result in a similar performance, whereas when the dataset with lumpy demand is considered, Croston's method leads to the best performance, followed by the new method.

Finally, it is important to note that the investigation conducted in this shows the need to reconsider the implementation of the Croston and SBA methods in enterprise resource planning type solutions and specialised forecasting software packages. In fact, when these two methods are applied in practice, a managerial intervention should be performed frequently to update the forecasts made by these methods and therefore to deal with the risk of obsolescence. Obviously, this is in line with the principle of the new developed methods that capture the risk of obsolescence through more frequent updates of the demand. This constitutes an interesting idea for future research.

Another interesting avenue for further research would be to derive the statistical properties of the modified SBA method, which enables a better understanding of the theoretical comparative performance of the methods. And yet another interesting avenue for research consists of broadening the empirical investigation by including other forecasting methods, such as the method developed by Prestwich et al. (2014), in order to gain more insights into the empirical comparative performance of the methods.

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Appendix A. ME and MSE detailed results for a very lumpy demand data

		ME						MSE					
	$\beta\alpha$	0.05	0.1	0.15	0.2	0.25	0.3	0.05	0.1	0.15	0.2	0.25	0.3
Croston	0.01	0.2002	0.1823	0.1694	0.1606	0.1549	0.1516	1.1343	1.1299	1.1289	1.1302	1.1333	1.1376
	0.02	0.2004	0.1825	0.1696	0.1607	0.1550	0.1516	1.1349	1.1304	1.1293	1.1306	1.1336	1.1378
	0.03	0.2008	0.1828	0.1698	0.1609	0.1551	0.1518	1.1356	1.1310	1.1298	1.1311	1.1340	1.1381
	0.04	0.2012	0.1832	0.1702	0.1612	0.1554	0.1520	1.1364	1.1317	1.1305	1.1316	1.1345	1.1385
	0.05	0.2018	0.1837	0.1707	0.1617	0.1558	0.1523	1.1373	1.1325	1.1312	1.1323	1.1350	1.1390
	0.1	0.2060	0.1876	0.1742	0.1649	0.1588	0.1550	1.1429	1.1374	1.1356	1.1363	1.1388	1.1424
	0.15	0.2120	0.1930	0.1793	0.1696	0.1632	0.1592	1.1502	1.1438	1.1414	1.1416	1.1437	1.1471
	0.2	0.2192	0.1997	0.1855	0.1755	0.1687	0.1645	1.1589	1.1514	1.1482	1.1480	1.1497	1.1529
	0.25	0.2274	0.2073	0.1926	0.1822	0.1751	0.1705	1.1688	1.1601	1.1560	1.1552	1.1566	1.1595
	0.3	0.2364	0.2157	0.2004	0.1896	0.1822	0.1773	1.1798	1.1697	1.1647	1.1633	1.1643	1.1669
SBA	0.01	0.1983	0.1805	0.1677	0.1589	0.1533	0.1500	1.1335	1.1292	1.1282	1.1296	1.1327	1.1370
	0.02	0.1967	0.1789	0.1661	0.1574	0.1517	0.1484	1.1334	1.1290	1.1280	1.1294	1.1323	1.1365
	0.03	0.1951	0.1774	0.1647	0.1559	0.1502	0.1468	1.1333	1.1290	1.1279	1.1292	1.1321	1.1361
	0.04	0.1937	0.1760	0.1633	0.1545	0.1488	0.1454	1.1333	1.1290	1.1279	1.1291	1.1319	1.1359
	0.05	0.1924	0.1747	0.1620	0.1532	0.1475	0.1441	1.1334	1.1291	1.1279	1.1291	1.1318	1.1357
	0.1	0.1869	0.1694	0.1567	0.1479	0.1420	0.1385	1.1349	1.1303	1.1289	1.1297	1.1321	1.1356
	0.15	0.1829	0.1654	0.1527	0.1437	0.1378	0.1341	1.1375	1.1325	1.1308	1.1313	1.1333	1.1364
	0.2	0.1797	0.1622	0.1494	0.1404	0.1343	0.1305	1.1407	1.1352	1.1331	1.1333	1.1350	1.1378
	0.25	0.1770	0.1594	0.1466	0.1374	0.1312	0.1273	1.1442	1.1383	1.1358	1.1356	1.1370	1.1395
	0.3	0.1746	0.1570	0.1440	0.1348	0.1285	0.1244	1.1479	1.1414	1.1385	1.1380	1.1392	1.1414
TSB	0.01	0.1581	0.1423	0.1309	0.1229	0.1176	0.1143	1.1335	1.1292	1.1282	1.1296	1.1327	1.1370
	0.02	0.1322	0.1177	0.1070	0.0994	0.0942	0.0908	1.1191	1.1164	1.1159	1.1172	1.1195	1.1227
	0.03	0.1151	0.1013	0.0910	0.0836	0.0784	0.0749	1.1149	1.1125	1.1121	1.1130	1.1148	1.1173
	0.04	0.1030	0.0897	0.0797	0.0723	0.0671	0.0635	1.1149	1.1124	1.1117	1.1123	1.1137	1.1158
	0.05	0.0942	0.0812	0.0713	0.0640	0.0588	0.0551	1.1169	1.1141	1.1131	1.1134	1.1145	1.1163
	0.1	0.0718	0.0595	0.0500	0.0428	0.0374	0.0335	1.1198	1.1167	1.1154	1.1153	1.1162	1.1177
	0.15	0.0627	0.0506	0.0413	0.0342	0.0288	0.0248	1.1383	1.1332	1.1304	1.1292	1.1293	1.1302
	0.2	0.0578	0.0459	0.0367	0.0295	0.0242	0.0202	1.1580	1.1509	1.1467	1.1446	1.1441	1.1448
	0.25	0.0548	0.0430	0.0338	0.0267	0.0213	0.0173	1.1785	1.1693	1.1637	1.1607	1.1597	1.1602
	0.3	0.0527	0.0410	0.0318	0.0247	0.0193	0.0153	1.2005	1.1892	1.1820	1.1780	1.1764	1.1767
Modified SBA	0.01	0.1536	0.1387	0.1278	0.1203	0.1154	0.1124	1.1137	1.1117	1.1119	1.1135	1.1162	1.1196
	0.02	0.1317	0.1180	0.1079	0.1009	0.0962	0.0933	1.1076	1.1063	1.1067	1.1082	1.1105	1.1135
	0.03	0.1176	0.1046	0.0950	0.0883	0.0837	0.0809	1.1046	1.1036	1.1040	1.1055	1.1077	1.1103
	0.04	0.1074	0.0949	0.0857	0.0792	0.0747	0.0719	1.1028	1.1020	1.1025	1.1039	1.1059	1.1084
	0.05	0.0995	0.0874	0.0785	0.0721	0.0678	0.0650	1.1016	1.1010	1.1015	1.1029	1.1048	1.1072
	0.1	0.0765	0.0655	0.0574	0.0515	0.0474	0.0447	1.0996	1.0993	1.0999	1.1011	1.1028	1.1047
	0.15	0.0647	0.0542	0.0464	0.0407	0.0368	0.0341	1.0997	1.0995	1.1000	1.1011	1.1026	1.1043
	0.2	0.0569	0.0467	0.0391	0.0336	0.0297	0.0271	1.1005	1.1002	1.1006	1.1016	1.1029	1.1045
	0.25	0.0512	0.0412	0.0337	0.0283	0.0244	0.0218	1.1016	1.1011	1.1014	1.1023	1.1035	1.1049
	0.3	0.0467	0.0368	0.0294	0.0240	0.0201	0.0175	1.1028	1.1022	1.1024	1.1031	1.1041	1.1054
SES		0.0542	0.0313	0.0219	0.0167	0.0134	0.0112	1.1261	1.1539	1.1832	1.2143	1.2474	1.2826

Table A1. Results for a linearly decreasing demand ($p_0 = 0.1$, $c = 0.9$) with a very lumpy pattern

Appendix B. Detailed empirical results on forecast accuracy

Estimator	α smoothing constant						β
	0.05	0.10	0.15	0.2	0.25	0.30	
Croston	0.580	0.577	0.574	0.572	0.571	0.570	0.01
	0.559	0.556	0.554	0.552	0.550	0.549	0.02
	0.541	0.538	0.535	0.533	0.532	0.531	0.03
	0.525	0.522	0.519	0.517	0.515	0.514	0.04
	0.511	0.508	0.505	0.503	0.501	0.500	0.05
	0.462	0.457	0.454	0.451	0.450	0.448	0.10
	0.435	0.430	0.427	0.424	0.422	0.420	0.15
	0.424	0.419	0.414	0.411	0.409	0.407	0.20
	0.424	0.418	0.413	0.410	0.407	0.405	0.25
	0.433	0.426	0.421	0.417	0.414	0.412	0.30
SBA	0.570	0.567	0.565	0.563	0.561	0.560	0.01
	0.540	0.537	0.535	0.533	0.531	0.530	0.02
	0.513	0.510	0.507	0.505	0.504	0.502	0.03
	0.488	0.485	0.482	0.480	0.478	0.477	0.04
	0.465	0.461	0.459	0.457	0.455	0.454	0.05
	0.371	0.367	0.364	0.362	0.360	0.359	0.10
	0.302	0.297	0.294	0.291	0.289	0.288	0.15
	0.247	0.242	0.239	0.236	0.234	0.232	0.20
	0.203	0.198	0.194	0.191	0.188	0.187	0.25
	0.166	0.161	0.156	0.153	0.150	0.149	0.30
TSB	-0.204	-0.200	-0.196	-0.194	-0.192	-0.190	0.01
	-0.187	-0.182	-0.178	-0.174	-0.172	-0.170	0.02
	-0.173	-0.167	-0.162	-0.158	-0.156	-0.153	0.03
	-0.161	-0.154	-0.149	-0.145	-0.142	-0.139	0.04
	-0.151	-0.144	-0.138	-0.134	-0.130	-0.127	0.05
	-0.119	-0.110	-0.103	-0.098	-0.093	-0.089	0.10
	-0.102	-0.093	-0.085	-0.079	-0.074	-0.070	0.15
	-0.092	-0.083	-0.075	-0.068	-0.062	-0.058	0.20
	-0.085	-0.076	-0.068	-0.061	-0.055	-0.050	0.25
	-0.081	-0.071	-0.063	-0.056	-0.050	-0.045	0.30
Modified SBA	-0.475	-0.472	-0.469	-0.467	-0.466	-0.465	0.01
	-0.391	-0.388	-0.385	-0.383	-0.382	-0.381	0.02
	-0.329	-0.326	-0.323	-0.321	-0.320	-0.318	0.03
	-0.280	-0.277	-0.274	-0.272	-0.270	-0.269	0.04
	-0.240	-0.237	-0.234	-0.232	-0.230	-0.229	0.05
	-0.108	-0.104	-0.101	-0.099	-0.097	-0.095	0.10
	-0.030	-0.026	-0.022	-0.020	-0.017	-0.016	0.15
	0.025	0.030	0.033	0.036	0.038	0.040	0.20
	0.067	0.072	0.076	0.079	0.081	0.083	0.25
	0.102	0.107	0.111	0.114	0.117	0.119	0.30
SES	0.118	0.071	0.048	0.035	0.028	0.022	

Table B1. ME results – dataset 1

Estimator	α smoothing constant						β
	0.05	0.10	0.15	0.2	0.25	0.30	
Croston	237.351	236.717	236.618	236.881	237.401	238.109	0.01
	236.743	236.114	236.007	236.254	236.751	237.433	0.02
	236.239	235.614	235.499	235.732	236.211	236.872	0.03
	235.817	235.194	235.073	235.294	235.758	236.400	0.04
	235.460	234.838	234.712	234.924	235.374	236.000	0.05
	234.330	233.705	233.559	233.738	234.145	234.723	0.10
	233.848	233.209	233.049	233.211	233.600	234.156	0.15
	233.739	233.078	232.906	233.060	233.443	233.994	0.20
	233.893	233.203	233.019	233.170	233.554	234.110	0.25
	234.267	233.539	233.341	233.490	233.880	234.447	0.30
SBA	237.202	236.574	236.474	236.733	237.247	237.947	0.01
	236.463	235.844	235.736	235.975	236.461	237.128	0.02
	235.841	235.230	235.114	235.338	235.800	236.438	0.03
	235.311	234.706	234.585	234.794	235.235	235.849	0.04
	234.855	234.256	234.130	234.326	234.749	235.341	0.05
	233.306	232.728	232.584	232.735	233.094	233.608	0.10
	232.462	231.896	231.743	231.867	232.188	232.653	0.15
	231.995	231.436	231.276	231.383	231.677	232.110	0.20
	231.762	231.205	231.039	231.134	231.409	231.818	0.25
	231.691	231.132	230.961	231.044	231.305	231.695	0.30
TSB	233.140	232.562	232.369	232.438	232.694	233.086	0.01
	233.028	232.414	232.202	232.258	232.501	232.880	0.02
	233.242	232.586	232.352	232.394	232.625	232.993	0.03
	233.634	232.938	232.682	232.710	232.932	233.293	0.04
	234.130	233.396	233.119	233.136	233.351	233.708	0.05
	237.236	236.343	235.983	235.963	236.169	236.536	0.10
	240.744	239.707	239.278	239.238	239.460	239.867	0.15
	244.474	243.287	242.792	242.741	242.990	243.453	0.20
	248.404	247.059	246.495	246.437	246.723	247.253	0.25
	252.537	251.021	250.387	250.327	250.656	251.264	0.30
Modified SBA	235.407	234.853	234.771	235.010	235.473	236.103	0.01
	234.082	233.562	233.475	233.681	234.093	234.658	0.02
	233.266	232.764	232.668	232.848	233.223	233.741	0.03
	232.707	232.216	232.112	232.271	232.615	233.097	0.04
	232.297	231.813	231.702	231.843	232.163	232.614	0.05
	231.243	230.775	230.635	230.718	230.957	231.312	0.10
	230.834	230.372	230.217	230.269	230.464	230.765	0.15
	230.662	230.205	230.042	230.075	230.243	230.512	0.20
	230.613	230.159	229.993	230.014	230.164	230.409	0.25
	230.652	230.199	230.029	230.040	230.177	230.404	0.30
SES	233.804	239.242	245.528	252.291	259.472	267.066	

Table B2. MSE results – dataset 1

Estimator	α smoothing constant						β
	0.05	0.10	0.15	0.2	0.25	0.30	
Croston	0.112	0.119	0.123	0.126	0.128	0.128	0.01
	0.110	0.117	0.122	0.125	0.126	0.127	0.02
	0.109	0.116	0.121	0.124	0.126	0.127	0.03
	0.108	0.116	0.120	0.124	0.125	0.127	0.04
	0.108	0.116	0.121	0.124	0.126	0.127	0.05
	0.112	0.120	0.126	0.130	0.132	0.134	0.10
	0.121	0.130	0.136	0.140	0.143	0.145	0.15
	0.132	0.142	0.149	0.153	0.157	0.159	0.20
	0.146	0.156	0.163	0.168	0.171	0.173	0.25
	0.159	0.170	0.177	0.183	0.186	0.189	0.30
SBA	0.090	0.096	0.101	0.103	0.105	0.106	0.01
	0.065	0.072	0.076	0.079	0.081	0.082	0.02
	0.041	0.048	0.053	0.056	0.057	0.058	0.03
	0.018	0.025	0.030	0.033	0.034	0.036	0.04
	-0.005	0.002	0.007	0.010	0.012	0.013	0.05
	-0.115	-0.107	-0.102	-0.098	-0.096	-0.094	0.10
	-0.220	-0.212	-0.206	-0.202	-0.199	-0.197	0.15
	-0.323	-0.314	-0.308	-0.304	-0.301	-0.299	0.20
	-0.425	-0.416	-0.410	-0.406	-0.403	-0.401	0.25
	-0.528	-0.519	-0.512	-0.508	-0.505	-0.503	0.30
TSB	0.061	0.068	0.072	0.075	0.076	0.077	0.01
	0.060	0.067	0.072	0.074	0.076	0.077	0.02
	0.059	0.066	0.071	0.074	0.076	0.077	0.03
	0.057	0.065	0.070	0.073	0.075	0.077	0.04
	0.056	0.064	0.069	0.073	0.075	0.076	0.05
	0.049	0.058	0.064	0.068	0.070	0.072	0.10
	0.042	0.051	0.058	0.062	0.065	0.067	0.15
	0.034	0.044	0.051	0.055	0.058	0.060	0.20
	0.027	0.037	0.044	0.049	0.052	0.054	0.25
	0.020	0.031	0.038	0.043	0.046	0.048	0.30
Modified SBA	0.085	0.092	0.097	0.099	0.101	0.102	0.01
	0.057	0.064	0.069	0.071	0.073	0.074	0.02
	0.030	0.037	0.042	0.045	0.046	0.048	0.03
	0.004	0.011	0.016	0.019	0.021	0.022	0.04
	-0.022	-0.014	-0.010	-0.006	-0.004	-0.003	0.05
	-0.142	-0.134	-0.128	-0.125	-0.122	-0.121	0.10
	-0.254	-0.245	-0.240	-0.236	-0.233	-0.231	0.15
	-0.362	-0.353	-0.347	-0.343	-0.340	-0.338	0.20
	-0.468	-0.459	-0.453	-0.449	-0.446	-0.444	0.25
	-0.572	-0.564	-0.558	-0.553	-0.550	-0.548	0.30
SES	0.066	0.068	0.066	0.062	0.057	0.052	

Table B3. ME results – dataset 2

Estimator	α smoothing constant						β
	0.05	0.10	0.15	0.2	0.25	0.30	
Croston	75.255	75.211	75.732	76.603	77.709	78.989	0.01
	75.220	75.178	75.700	76.573	77.682	78.965	0.02
	75.196	75.155	75.680	76.555	77.667	78.952	0.03
	75.180	75.142	75.669	76.547	77.661	78.949	0.04
	75.173	75.137	75.666	76.547	77.664	78.955	0.05
	75.222	75.199	75.743	76.639	77.773	79.081	0.10
	75.365	75.356	75.917	76.831	77.982	79.308	0.15
	75.561	75.567	76.145	77.077	78.246	79.590	0.20
	75.788	75.810	76.406	77.355	78.542	79.904	0.25
	76.037	76.075	76.687	77.655	78.859	80.239	0.30
SBA	75.214	75.166	75.678	76.537	77.630	78.895	0.01
	75.142	75.092	75.598	76.447	77.529	78.781	0.02
	75.086	75.034	75.532	76.373	77.443	78.683	0.03
	75.042	74.989	75.481	76.312	77.371	78.598	0.04
	75.010	74.955	75.441	76.263	77.311	78.527	0.05
	74.985	74.923	75.378	76.157	77.153	78.309	0.10
	75.122	75.054	75.481	76.217	77.162	78.260	0.15
	75.376	75.302	75.701	76.396	77.289	78.329	0.20
	75.726	75.645	76.016	76.669	77.512	78.494	0.25
	76.162	76.074	76.417	77.029	77.821	78.747	0.30
TSB	75.192	75.156	75.679	76.547	77.649	78.921	0.01
	75.141	75.104	75.626	76.494	77.595	78.867	0.02
	75.105	75.068	75.589	76.457	77.557	78.830	0.03
	75.081	75.044	75.566	76.433	77.534	78.806	0.04
	75.069	75.033	75.554	76.422	77.523	78.795	0.05
	75.131	75.102	75.629	76.500	77.603	78.877	0.10
	75.331	75.313	75.849	76.727	77.836	79.114	0.15
	75.612	75.610	76.157	77.045	78.161	79.446	0.20
	75.951	75.964	76.526	77.426	78.553	79.847	0.25
	76.333	76.364	76.942	77.856	78.996	80.302	0.30
Modified SBA	75.204	75.155	75.666	76.523	77.614	78.877	0.01
	75.127	75.075	75.577	76.423	77.501	78.749	0.02
	75.067	75.012	75.505	76.341	77.406	78.640	0.03
	75.021	74.963	75.449	76.274	77.327	78.548	0.04
	74.988	74.928	75.407	76.221	77.261	78.469	0.05
	74.970	74.899	75.343	76.109	77.091	78.234	0.10
	75.122	75.043	75.455	76.175	77.102	78.183	0.15
	75.393	75.306	75.688	76.363	77.236	78.256	0.20
	75.759	75.664	76.017	76.648	77.469	78.429	0.25
	76.210	76.107	76.431	77.020	77.789	78.691	0.30
SES	74.970	75.269	76.339	77.859	79.673	81.702	

Table B4. MSE results – dataset 2